

Focus of this homework:

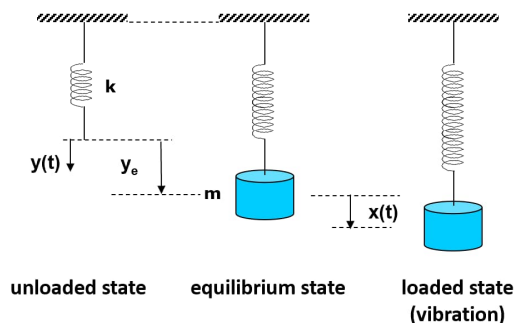
Fundamentals of undamped and damped free vibration of SDOF vibratory systems (modeling, application of the energy conservation principle for undamped systems, and time response).

Submit homework (in hard copies) by the beginning of the class. It is your responsibility to make sure that homework is submitted in good quality. Please organize your work neatly. Suggest to start each problem on a new page.

1) Review of general concepts of physics:

- Estimate the suspension stiffness and the frequency of rotation of a tire of a passenger vehicle at a given speed. Make reasonable estimations or simple measurements as needed.
- Describe a real-world example of vibration system and its response characteristics.
- The following example shows how the derivation of the governing equation of motion can differ based on the choice of an absolute coordinate $y(t)$ versus coordinate $x(t)$ measured from the equilibrium position. Discuss the important lessons learned here.

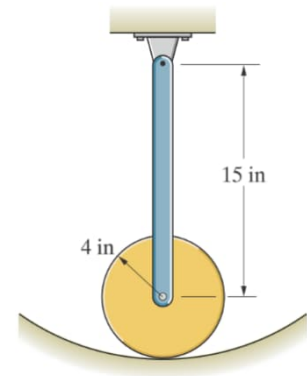
A Prototypical Vibratory System



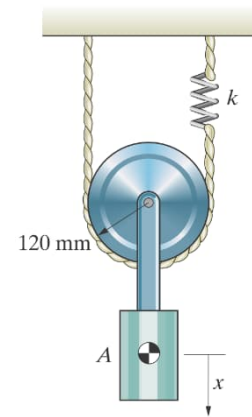
$$\begin{aligned} m\ddot{y} &= mg - F_s, \text{ where } F_s = ky \\ \Rightarrow m\ddot{y} &= mg - k(y_e + x) \\ mg &= ky_e, \quad y = y_e + x, \quad \ddot{y} = \ddot{x} \\ \Rightarrow m\ddot{x} + kx &= 0 \text{ or } m_{eq}\ddot{x} + k_{eq}x = 0 \end{aligned}$$

This is the basic form of governing equation for the free vibration of SDOF undamped systems

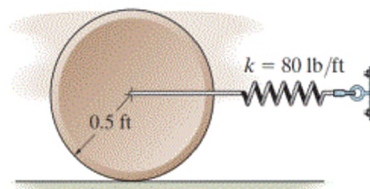
- 2) The 2-lb bar is pinned to the 5-lb disk. The disk rolls on the circular surface. What is the frequency of small vibrations of the system relative to its vertical equilibrium position?
 Hint: use energy method.



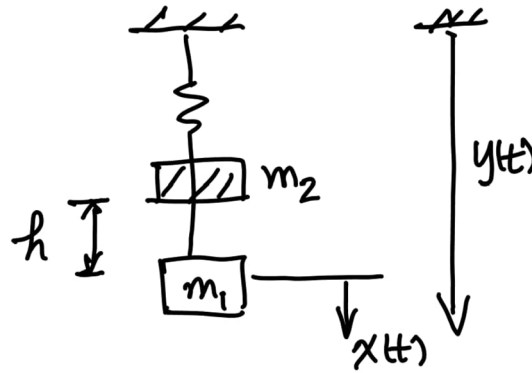
- 3) The mass of the suspended object A is 4 kg. The mass of the pulley is 2 kg and its moment of inertia is 0.018 kg-m^2 . The spring constant $k = 150 \text{ N/m}$. For vibration of the system relative to its equilibrium position, determine (a) the frequency in Hz, and (b) the period of oscillation.



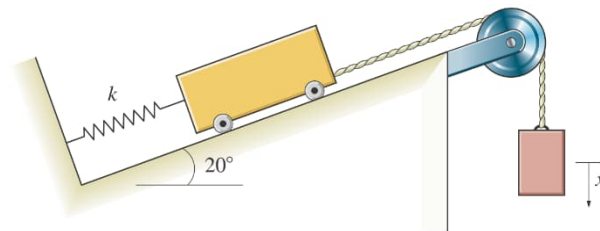
- 4) The disk has a weight of 30 lb and rolls without slipping on the horizontal surface as it oscillates about its equilibrium position. If the disk is displaced, by initially rolling it counterclockwise 0.2 rad and rotating it clockwise 1 rad/s, determine the equation which describes its oscillatory motion and the natural period when it is released. Also determine the response solution and the maximum angular velocity during the ensuing motion.



- 5) A mass m_1 is in static equilibrium position when a mass m_2 is dropped a height of h and impacts m_1 . After impact, there is no rebound (i.e., the coefficient of restitution $e = 0$). Solve for the response using the two different coordinate systems discussed in class. (Please refer to the lecture notes [for review](#).)



- 6) The spring, with $k = 800 \text{ N/m}$, is unstretched when $x = 0$. The mass of each object is 30 kg . The inclined surface is smooth. The radius of the pulley is 120 mm and its moment of inertia about its center is $I = 0.03 \text{ kg}\cdot\text{m}^2$. At $t = 0$, $x = 0$ and $dx/dt = 1 \text{ m/s}$.
- Determine the frequency and period of the resulting vibration.
 - What is the value of the vibration response $x(t)$ at $t = 4$ seconds?



- 7) Consider the damped oscillation of the system with the spring unstretched in the position shown (the massless rigid link is hinged on the left end). The values of the parameters are: $m = 1,750$ kg, $c = 3,500$ N-s/m, $k = 7 \times 10^5$ N/m, $a = 1.25$ m, and $b = 2.5$ m. Let $\theta(t)$ be the angle measured from the undeformed configuration. You can remove the effect of the gravity term by simply considering this system on a horizontal plane.
- Derive the governing equation of motion with respect to the equilibrium. Calculate the damped natural frequency of the system
 - Determine the solution for the response of this system. The initial conditions of the system are: $\theta(0) = 0.1$ rad, $\dot{\theta}(0) = 0$.
 - Determine the value of the critical damping and numerically simulate the response.

